INCH-POUND
MIL-PRF-62494D(AT)
30 September 1998
SUPERSEDING
MIL-PRF-62494C(AT)
26 October 1995

PERFORMANCE SPECIFICATION

SHOCK ABSORBER, DIRECT ACTING

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, test, and manufacture requirements for the direct acting shock absorber for use on military tracked vehicles in the Bradley Fighting Vehicle (BFV) Family.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2540

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2000 - Rubber Products in Automotive Applications (DoD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Materials, design, and construction. Unless otherwise specified herein, the design, materials and construction process selection. is the prerogative of the contractor as long as all articles submitted to the Government fully meet the operating, interface, support and ownership, and operating environment requirements specified. Shock absorbers shall conform to the manufacturer's designs and construction except for the tolerances, envelope and mounting dimensions which shall conform to figures 1, 2 and 3 (see 4.3.2).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.3.2).

3.2.2 Ozone and petroleum resistance. Rubber components under stress in usage and/or subject to ozone attack which causes impairment of function and durability, shall be ozone resistant and shall conform to ASTM D2000, C12 requirements. Rubber components shall be resistant to petroleum oils and lubricants, servicing fluids where their intended use requires it (see 4.3.2).

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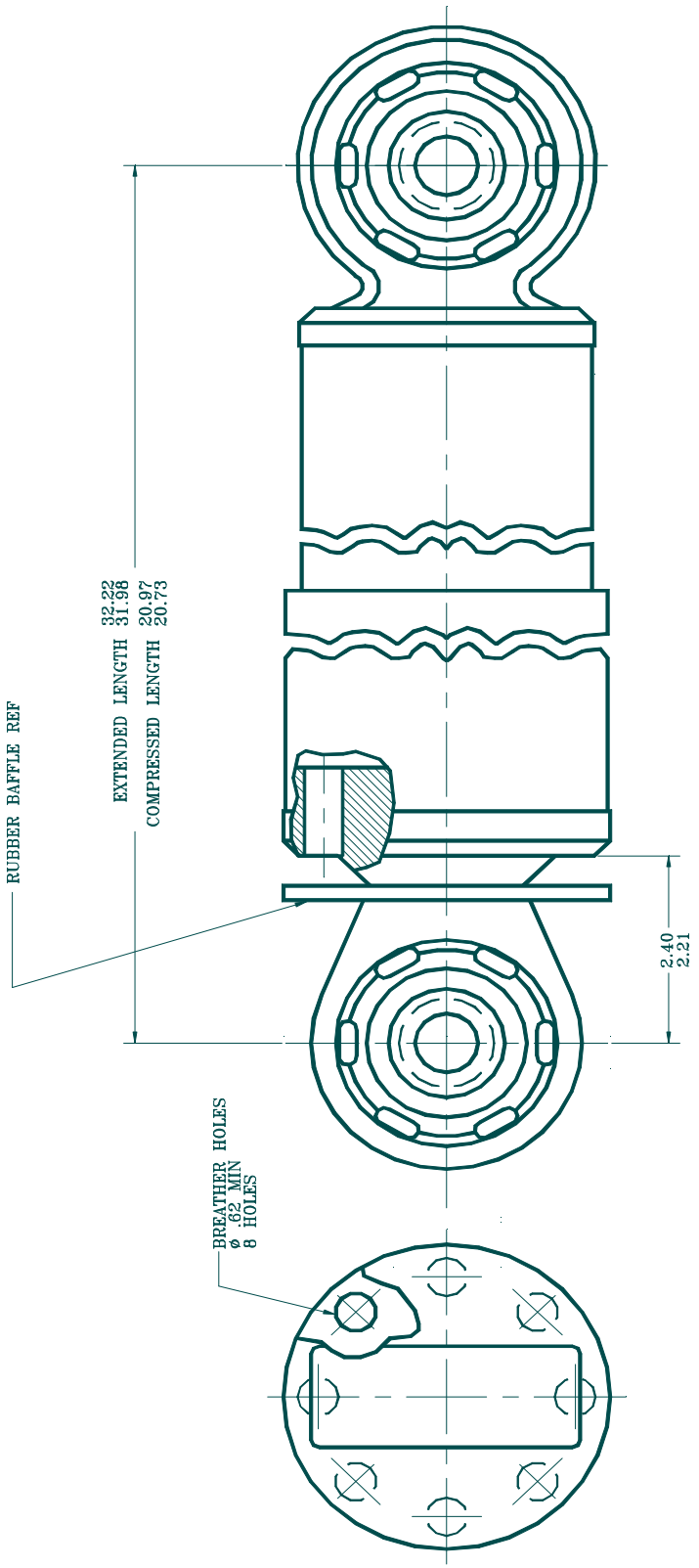


FIGURE 1. Side view shock absorber.

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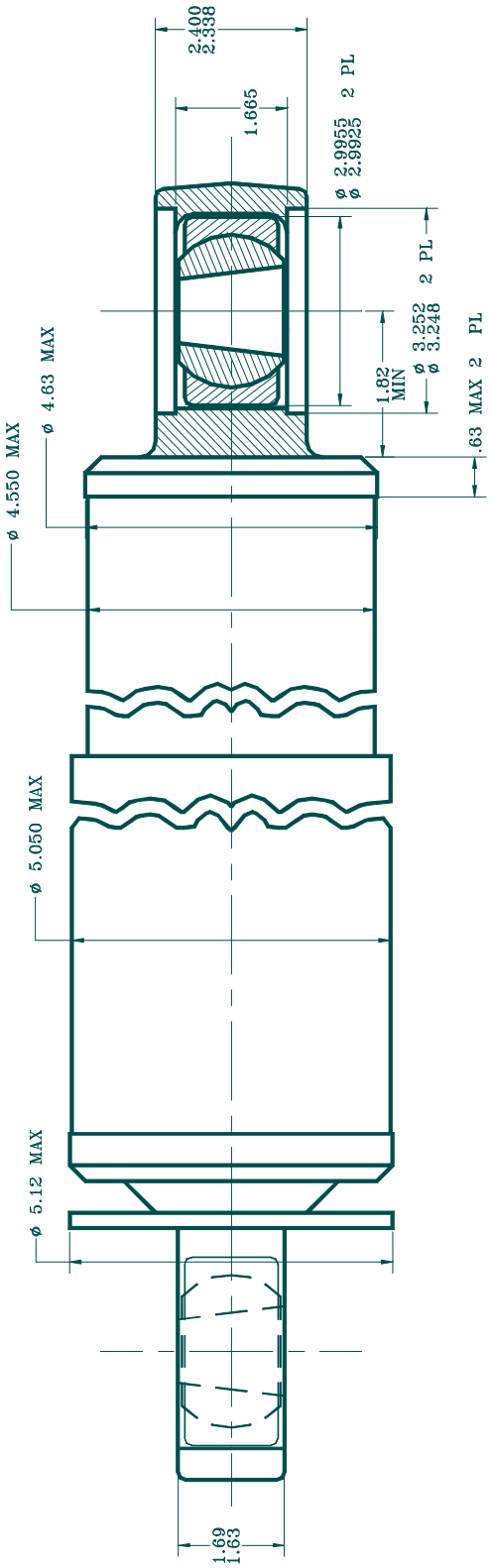


FIGURE 2. Top view shock absorber.

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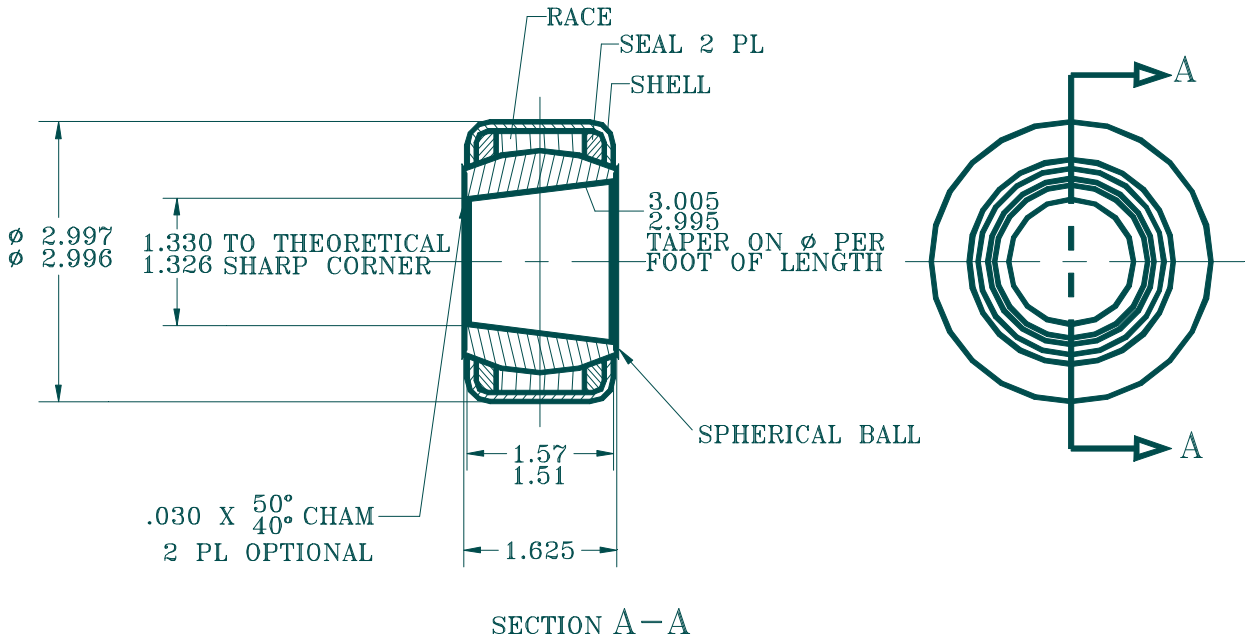


FIGURE 3. Bearings.

3.2.3 Bearing seats. The internal bearing seats between the ball and the outer bearing shell shall be self-aligning, permanently lubricated, and be of a resilient nature to prevent brinelling (see 4.3.2).

3.2.4 Thermal design. All components shall be of material and design sufficient to withstand intermittent operation at an oil temperature of 580 degrees Fahrenheit (°F) (305 degrees Celsius (°C)) (see 4.3.2).

3.3 Operating requirements. Each shock absorber shall provide the following functional, operational, and performance capabilities.

3.3.1 Damping resistance. Damping resistance shall be within the limits specified in table I (see 4.4.1).

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TABLE I. Damping resistance.

Performance data sinusoidal motion 3-inch (7.62 cm) stroke				
Cycling rate (cpm) 1/	Resistance at midstroke			
	Compression		Rebound	
	pounds	(kg)	pounds	(kg)
47-53	3300-6800	(1497 to 3084)	2400-3300	(1089 to 1497)
94-106	5100-7100	(2313 to 3220)	2500-3600	(1134 to 1633)
141-159	5200-7300	(2359 to 3311)	2700-3900	(1225 to 1769)
282-318	5600-8200	(2540 to 3719)	3100-4700	(1406 to 2132)
564-636	6400-11 100	(2903 to 5035)	3400-9400	(1542 to 4264)

1/ cpm = cycles per minute

NOTES:

1. Resistance at first 10 percent of each stroke shall be at least 1650 pounds (lb) (748 kilograms (kg)) in compression and 1200 lb (544 kg) in rebound at all cycling speeds noted above.
2. Limits apply for shock absorber temperature within ranges of 50° to 100°F (10° to 38°C) and 350° to 400°F (177° to 204°C).

3.3.2 Compression and tension loading. The shock absorber shall be capable of withstanding a 40 000 lb (18 144 kg) compressive force and a 40 000 lb tensile force, applied through the self-aligning bearings, without affecting the damping capability (see 4.4.2).

3.3.3 Endurance. The shock absorber shall be capable of operating satisfactorily for 300 000 feet (ft) (91 440 meter (m)) of piston travel at a standard operational temperature range of 300° to 400°F (149° to 204°C). The shock absorber shall be capable of withstanding piston velocities of 100 inches per second (2.54 meter per second (m/s)) at mid-stroke in compression and rebound. There shall be minimal loss of damping, fluid and performance (see 4.4.3).

3.3.4 Bearing rotational torque. The torque required to rotate the upper, spherical inner race at 3 to 5 revolutions per minute (rpm) shall be from 12 to 100 foot-pounds (ft-lb), (16.7 to 135.6 Newton-meter (N-m)). The torque required to rotate the lower spherical inner race at 3 to 5 rpm shall be from 8 to 150 ft-lb (10.8 to 203 N-m) (see 4.4.4).

3.3.5 Axial load. The self-aligning bearing shall not move from its installed location nor shall it be permanently deformed when an axial load of 2000 lb (907 kg) is applied in either direction to the inner bearing shell (see 4.4.5).

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3.3.6 Angular velocity. The self-aligning bearing shall not degrade overall performance below acceptable limits and shall retain its unit integrity when subjected to a radial load of 6000 lb (2722 kg) and when the inner race is oscillated through a 65 ±5 degree angle at a rate of 1 ±0.2 cycles per second for 1 hour (see 4.4.6).

3.4 Interface requirements. Each shock absorber shall accommodate the following inputs and interfaces.

3.4.1 Vehicle system interface. The shock absorber shall be compatible with the mating vehicle suspension as stated in the contract or order, with minimal loss of performance (see 4.5.1 and 6.2).

3.4.2 Interchangeability. All shock absorbers shall be interchangeable as to fit, form, and function with other manufacturer's assemblies having the same military part number (see 4.5.2).

3.5 Support and ownership requirements. Each shock absorber shall possess the following life cycle ownership characteristics.

3.5.1 Bearing. The self-aligning ball bearing shall be fabricated from materials that are inherently corrosion resistant or are permanently treated to prevent forms of corrosion and deterioration which affect the bearing's performance. The balls shall be hardened to Rockwell C40 minimum (see 4.6.1).

3.5.2 Reliability requirements. The shock absorber shall demonstrate a minimum average life of 5000 miles (8046 kilometers (km)) during vehicle testing (see 4.6.2).

3.5.3 Reserve tube durability. The reserve tube shall have a minimum yield of 60 000 pounds per square inch (psi) (414 kilopascal (kPa)) and wall thickness to prevent denting (see 4.6.3).

3.5.4 Dirt shield tube durability. The dirt shield tube shall have a minimum yield of 60 000 psi (414 kPa) and wall thickness to prevent denting (see 4.6.4).

3.5.5 Protective finish and color. Unless otherwise specified, (see 6.2) the shock absorber shall be cleaned, pretreated and primed in accordance with the manufacturer's standard practice. Unless otherwise specified, (see 6.2) the topcoat color shall be forest green, Chemical Agent Resisting Coating (CARC) (see 6.5). Painting shall be accomplished without overspray on the piston rod surfaces (see 4.6.5).

3.5.6 Identification marking. Identification marking shall include, as a minimum, the following information permanently marked on the exterior of the shock absorber (see 4.6.6 and 6.2):

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a. Manufacturer's identification
b. Serial number
c. Military part number
d. National stock number
e. Manufacturer's part number
f. Date of manufacture
g. Contract number
h. U.S.

3.5.7 Safety locked piston nut. Where applicable, after torquing, the piston nut shall be secured, by appropriate means, to prevent loosening or separation of the shock absorber from the mounted position. The shock absorber shall not separate from the mounting, to prevent endangering personnel or damage to equipment (see 4.6.7).

3.5.8 Weight. Weight of the shock absorber shall be not greater than 50 lb (22.7 kg) (see 4.6.8).

3.6 Operating environment requirements. Each shock absorber shall operate under the following environmental conditions, without degradation in performance.

3.6.1 Low temperature. The shock absorber shall demonstrate no performance degradation and show no evidence of damage or deformation after being stored for not less than 8 hours at -70°F (-57°C). Fluid leakage shall be not greater than 1 drop per hour (see 4.7.1).

3.6.2 Steam and waterjet cleaning. The shock absorber shall demonstrate no performance degradation and show no evidence of damage or deformation after being subjected to steam cleaning, followed by waterjet cleaning (see 4.7.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspections (see 4.1.2).

4.1.1 First article inspection. Unless otherwise specified, (see 6.2 and 6.3), a first article inspection shall be performed when a first article sample is required (see 3.1). First article sample inspection shall consist of all the verifications as listed in table II (see 4.3 and 4.3.1). Failure of any inspection shall be cause for rejection.

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4.1.2 Conformance inspections. Conformance inspection shall include the examinations and tests from table II, as defined in the contract (see 6.2).

Table II. Verification methods.

Title	Requirement	Verification
Materials, design and construction	3.2	4.3.2
Operating requirements	3.3	4.4
Damping resistance	3.3.1	4.4.1
Compression and tension loading	3.3.2	4.4.2
Tension loading endurance	3.3.3	4.4.3
Bearing rotational torque	3.3.4	4.4.4
Axial load	3.3.5	4.4.5
Angular velocity	3.3.6	4.4.6
Interface requirements	3.4	4.5
Vehicle system testing	3.4.1	4.5.1
Interchangeability	3.4.2	4.5.2
Support and ownership requirements	3.5	4.6
Bearing	3.5.1	4.6.1
Reliability	3.5.2	4.6.2
Reserve tube durability	3.5.3	4.6.3
Dirt shield tube durability	3.5.4	4.6.4
Protective finish and color	3.5.5	4.6.5
Identification marking	3.5.6	4.6.6
Safety locked piston nut	3.5.7	4.6.7
Weight	3.5.8	4.6.8
Operating environment requirements	3.6	4.7
Low temperature	3.6.1	4.7.1
Steam and waterjet cleaning	3.6.2	4.7.2

4.2 Order of inspection. The inspection sequence may be in any order except that operation test shall be last (see 6.2).

4.3 Verification methods. The types of verification methods included in this section are visual, inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives to replace verifications required by this specification.

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4.3.2 Materials, design, and construction. Conformance to 3.2 through 3.2.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements.

4.3.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F (25° ± 10°C)
b. Relative humidity: uncontrolled room ambient
c. Atmospheric pressure: site pressure.

4.4 Operating requirements verification.

4.4.1 Damping resistance. To determine conformance to 3.3.1, the shock absorber shall be mounted in a suitable test fixture by means of the self-aligning bearings and shall be cycle tested in rebound and compression. During testing, the compression and rebound damping resistance at various piston velocities shall be measured. The number of measurements taken shall be of sufficient quantity to establish that the damping resistance to the shock absorber is within the limits specified in table I. Temperatures shall be measured on the surface of the shock absorber at the lower end of the reservoir. Cycle rates and temperature ranges for first article, shall be as specified.

4.4.2 Compression and tension loading. To determine conformance to 3.3.2, the shock absorber shall be mounted in a suitable test fixture by means of self-aligning bearings and shall be loaded with not less than 40 000 lb (18 144 kg) in compression and not less than 40 000 lb (18 144 kg) in tension without damage to any of its components. Damage to any of the components shall constitute failure of the test.

4.4.3 Endurance. To determine conformance to 3.3.3, the shock absorber shall be mounted in a suitable test fixture by means of the self-aligning bearings and shall be tested for not less than 600 000 cycles using a 3 ±0.125 in. (76 ±3 mm) stroke at a piston velocity necessary to produce a continuous fluid reservoir surface temperature of 350° to 400°F (177° to 204°C). External cooling may be provided as required. External leakage during testing shall be no greater than 15 percent of the reservoir fluid when the shock absorber is fully collapsed. Following testing no more than 25 percent change in damping resistance as measured prior to the start of testing shall constitute failure. When the damping resistance measurements are taken, the fluid reservoir surface temperature shall be 300° to 400°F (149° to 204°C).

4.4.4 Bearing rotational torque. Using any method for measuring bearing rotational torque (see 4.3), verify that the torque required to rotate the upper and lower spherical races at 3 to 5 rpm conforms to the values specified in 3.3.4. Torque above or below these specified values shall constitute failure.

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4.4.5 Axial load. To determine conformance to 3.3.5, an axial load through the bearing of 2000 lb (907 kg) shall be applied to the inner bearing shell, first in one direction and then in the opposite direction. Following testing, movement of the self-aligning bearing from its installed location or permanent deformation shall constitute failure.

4.4.6 Angular velocity. To determine conformance to 3.3.6, a radial load of 6000 lb (2722 kg) shall be applied to the self-aligning bearing. Bearing shall stay in its installed location. One cycle equals 130 degrees of total angular motion.

4.5 Interface requirements verification.

4.5.1 Vehicle systems testing. Sixteen preproduction shock absorbers shall be used for vehicle testing. Eight shock absorbers shall be installed on each of two test vehicles which shall be operated for 6000 miles (9656 km) each and shall approximate conditions of initial production tests. Tests shall include 1260 miles (2028 km) of paved roads, 2700 miles (4345 km) of secondary roads and 2040 miles (3283 km) of cross country roads. Each shock absorber shall be baseline tested in accordance with 4.4.1 before vehicle testing to assure conformance to the requirements of 3.3.1. After completion of 6000 miles (9656 km), the damping resistance test specified in 4.4.1 shall be repeated. Performance more than 25 percent below the minimum requirements of 3.3.1 shall constitute failure of the test (see 3.4.1).

4.5.2. Interchangeability. To determine conformance to 3.4.2, use one or more of the methods outlined in 4.3 and 4.3.1 to verify that the shock absorbers are interchangeable with other manufacturer's shock absorber.

4.6 Support and ownership requirements verification.

4.6.1 Bearing. To determine conformance to 3.5.1 use one or more of the methods outlined in 4.3 and 4.3.1 to verify that the shock absorber meets the performance requirements.

4.6.2 Reliability requirements. The shock absorber shall meet the requirements specified in 3.5.2 during vehicle testing. The reliability calculation shall be based on the total accumulated shock absorber test miles before failure divided by the total number of shock absorbers tested. Failure is defined as shock absorber replacement required because of the inability to perform its required function. A shock absorber shall be considered a failure if one or more of the following conditions exist (see 3.5.2):

a. External leakage.
b. Little or no increase in surface temperature after cross country operation.
c. Severe bearing wear.
d. Loss of more than 25 percent of minimum damping resistance (see 3.3.1).

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If a shock absorber has operated for 6000 miles (9656 km) without replacement, its life shall be taken as 6000 miles for the reliability calculation.

4.6.3 Reserve tube durability. To determine conformance to 3.5.3, use one or more of the methods outlined in 4.3 and 4.3.1 to verify that the shock absorber meets the durability requirements.

4.6.4 Dirt shield tube durability. To determine conformance to 3.5.4, use one or more of the methods outlined in 4.3 and 4.3.1 to verify that the shock absorber meets the durability requirements.

4.6.5 Protective finish and color. To determine conformance to 3.5.5, use one or more of the methods outlined in 4.3 and 4.3.1 to verify that the shock absorber meets the finish and color requirements.

4.6.6 Identification marking. To determine conformance to 3.5.6, use one or more of the methods outlined in 4.3 and 4.3.1 to verify that the shock absorber meets the identification marking requirements.

4.6.7 Safety locked piston nut. To determine conformance to 3.5.7, use one or more of the methods outlined in 4.3 and 4.3.1 to verify that the shock absorber meets the piston nut requirements.

4.6.8 Weight. To determine conformance to 3.5.8, use one or more of the methods outlined in 4.3 and 4.3.1 to verify that the shock absorbers do not exceed the weight requirements.

4.7 Operating environment requirements verification.

4.7.1 Low temperature. To determine conformance to 3.6.1, the shock absorber shall be tested as follows:

a. Place shock absorber in a cold chamber for not less than 8 hours at -70°F (-57°C).
b. Remove shock absorber from cold test chamber and mount it in a suitable test fixture by means of the self-aligning bearings. Start cycle testing within 15 minutes after removal from the cold chamber. Cycle testing shall consist of operating the shock absorber through a 3 $\pm$0.125 in. (76 $\pm$3 mm) stroke for no less than 120 cycles within 30 minutes.

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NOTE: One cycle equals 6 in. (152 mm) and 120 cycles equal 60 ft (18.3 meters (m)) of piston travel.

c. Measure external oil leakage during testing.

Upon completion, any evidence of damage or deformation of the shock absorber, or any fluid leakage greater than one drop per hour shall constitute failure.

4.7.2 Steam and waterjet cleaning. To determine conformance to 3.6.2, the shock absorber shall be subjected to steam cleaning and a cleaner, followed immediately by waterjet cleaning as follows. The jet pressure shall be 105 ±5 pound-force per square inch gage (psig) (723 ±34 (kPa)) for steam cleaning, and 50 ±5 psig (344 ±34 kPa) for waterjet cleaning. Prior to and within 1 hours (unless otherwise directed by the Government) after completing the steam and waterjet cleaning, the shock absorber shall be subjected to the damping resistance test specified in 4.4.1. There shall be no entry of water or moisture to modules sealed against water or moisture. Moisture inside the shock absorber shall not be considered a failure. Failure of the shock absorber to meet the requirements of the damping resistance test shall constitute failure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Shock absorbers manufactured in accordance with this specification are intended for use on military tracked vehicles in the BFV family. The shock absorbers are military unique as they must operate satisfactorily under adverse terrain conditions and extremes of temperature typically found on the battlefield. There is no known commercial equivalent that satisfies the requirements.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
c. When first article is required (see 3.1 and 4.1.1).
d. Applicable mating vehicle suspension (see 3.4.1)
e. If protective finish and the topcoat color of the CARC paint are other than as specified (see 3.5.5).
f. Identification and marking if other than as specified (see 3.5.6).
g. Conformance inspection as specified in the contract (see 4.1.2).
h. If order of inspection is required (see 4.2).
i. Inspection conditions if other than as specified (see 4.3.3).
j. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Presolicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from Table II and their designated frequency based on a risk assessment for the procurement.

6.5 Protective finish. Historically, CARC paint in accordance with MIL-C-46168 has been provided on this item.

6.5.1 Color. Historically, color chip 383 of FED-STD-595 has provided a close match to the required color.

6.6 Specification conformance. Historically, shock absorbers acquired to this document conform to specification 12369332. The bearings conform to Drawing 12359649.

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6.7 Definitions.

6.7.1 Direct acting shock absorber. The shock absorber is a self-contained, double-acting, energy-absorbing, hydraulic unit. The top of the shield has ventilation holes and is covered with a flexible rubber baffle to limit debris entrance through the holes and to permit the exit of mud. Compression and rebound valves are contained in the piston which is fastened to the actuation rod. One end of the innermost cylinder is formed by the base valve, which contains a compressive and an intake valve. Mounting of the shock absorber is provided by two self-aligning bearings, one at each end.

6.8 Subject term (key word) listing.

Axial load
Bearing
Compression
Cycling rate
Damper
Damping resistance
Torque
Valve spring

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2540-0456)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-62494D(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 980930

3. DOCUMENT TITLE SHOCK ABSORBER, DIRECT ACTING

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** (810) 574-8745 **(2) AUTOVON** 786-8745
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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